Daily Drilling Report
SPD19B-03
8/21/2014

PETROPARS Oil & Gas Developer

N.I.O.C. Pars Oil & Gas Company

Well Data

Well Name	Operator	Field Name	Well Type 1	Rig	Code
SPD19B-03	Petropars Ltd.	South Pars Gas Field	Development	Deep Driller, DD6	SP19-3B-01-DL-FOR-0...
Report Number	**Hole Section**	**Days LTI (days)**	**Water Depth (m)**	**Original KB Elevation (m)**	**Head Count**
25.0	12 1/4"	170.00	72.83	31.70	96.0
Last Cas	**Daily Cost Total**	**Daily Mud Cost**	**Cum Cost To Date**	**AFE Number**	**Cum Time Log Days (days)**
13 3/8" Shoe, 1,487.00mKB	44,333.81	285.60	5,527,863.47		21.85
Depth Start (mKB)	**Depth End (mKB)**	**Depth End (TVD) (mKB)**	**Depth Progress (m)**	**Drilling Hours (hrs)**	**Average ROP (m/hr)**
1,492.00	1,495.00	1,427.34	3.00	1.00	3.0

Operations at Report Time
Drilling @ 1533m.

Operations Summary
Cont' Slip & Cut. Resume RIH rotary BHA f/ 1100m and D/O Shoe track and rat hole f/1461.8m to 1492m. Drill 3m new formation to 1495m. Perform FIT, Ok. POOH to surface while L/D high rotating hours DP. RIH directional BHA#8 to 1000m.

Operations next 24hrs
Cont' RIH to Btm and directional drilling 12 1/4" hole section.

Supervisors Contact

Job Contact	Position
Mojtaba Parvaresh	Drilling Supt.
Sajad Tamaddon	Senior Drilling Sup.
Jabbar Mohammadi	Night Drilling Sup.
Hamed Daneshkhah / B.Khalilmoghadam	Drilling Engineer
Ali Sahebnouri	Geologist
R. Kamalpour	POGC Rep.

HSE Data

Type	Last Date	Next Date
Abandon Drill	8/10/2014	9/30/2014
BOP Test	8/20/2014	9/11/2014
Chemical Spill Drill	8/17/2014	10/17/2014
Fire Drill	8/10/2014	9/30/2014
H2S Drill	8/16/2014	9/16/2014
Heli Crash Drill	7/15/2014	9/14/2014
Medical Drill	7/26/2014	9/25/2014
MOB Drill	4/16/2014	6/16/2014
Pit Drill	8/2/2014	9/17/2014
Trip Drill	8/21/2014	10/11/2014

STOP Cards

Number of Reports: 36

Operations Summary

Start Time	End Time	Dur (hrs)	Phase	Code 1	Code 2	Time P-N-T	Comment
00:00	01:00	1.00	12 1/4"	D	20	P	Cont' Slip & Cut 110 ft drill line.
01:00	03:00	2.00	12 1/4"	D	04	P	Cont' RIH rotary BHA#7 on 5 7/8" DP f/ 1100m to 1435m. Wash down to 1461.8m. Tag CMT Plug w/ 10 kips.
03:00	06:00	3.00	12 1/4"	D	03	P	D/O cement plugs @ 1461.8 m + Float Collar @ 1462.8m + cement inside shoe track + Float Shoe @ 1486.7m and clear rathole to 1492m w/ 300-900 gpm/ 200-1100 psi/ 5-30 kips/ 50 rpm/ 9-16 kft-lbs. * Pump 30 bbls Hi-Vis pill after drill out Float Collar and middle of shoe track. * Reciprocate string inside float collar and shoe with and without rotation, Ok. * Start displacing at 1483m: Pump 30 bbls Hi-Vis + 50 bbls SW and displace hole to 9.1 ppg Polymer Mud.
06:00	07:00	1.00	12 1/4"	D	03	P	Drill new formation f/ 1492m to 1495m w/ 800-900 gpm/ 780-1100 psi/ 60 rpm/ 30 Kips/ 7-14 kft-lbs, meanwhile displacing hole, MW in & out : 9.1 ppg.
07:00	07:30	0.50	12 1/4"	D	11	P	Pull string inside CSG (1478m). L/U CMT unit. to string and annulus. Flush and pressure test surface lines to 2000 psi/ 5 min, Ok. Close top Pipe Ram. Perform FIT to 14 ppg EMW w/ 9.1 ppg mud by applying 1200 psi surface pressure in steps for 10 min, Ok. Pump totally 7.7 bbls and get same in return.
07:30	16:00	8.50	12 1/4"	D	04	P	POOH rotary BHA f/ 1478m to 549m while L/D high rotating hours 5 7/8" DP jt by jt.
16:00	18:00	2.00	12 1/4"	D	04	P	Cont' POOH f/ 549m to surface while R/B new 5 7/8" DP as Std in the derrick. B/O Bit (1-1-NO-A-E-I-NO-BHA) and R/B the rest of BHA. ** Make totally 13 new std and L/D 114 jt of high rotating hours DP.
18:00	20:30	2.50	12 1/4"	D	04	P	P/U and M/U directional BHA#8 consists of 12 1/4" BEST PDC Bit (Type: M1385, IADC: M432, SN: 228378, Nozzles: 4 x 12 + 4 x 13) + 9 5/8" Mud Motor (AKO: 1.00° , Sleeve Stab: 12 1/8") + 12" String Stabilizer (COR3059AD) + 8" NMDC + 8" TM Sub) Load Gamma and MWD probes inside). RIH to 23m. Align Motor with MWD.
20:30	21:00	0.50	12 1/4"	D	12	P	Connect TDS and perform Motor & MWD shallow hole test w/ 800 gpm/ 780 psi , OK.
21:00	22:00	1.00	12 1/4"	D	04	P	Cont' M/U rest of BHA items consists of: 8" NMDC + 8 1/2" Float sub with float valve inside + 1 jt 8 1/4" DC + X/O + 16 jts 5 7/8" HWDP + 7" Drilling Jar (COR3022AD) + 3 jts 5 7/8" HWDP and RIH to 226m.
22:00	00:00	2.00	12 1/4"	D	04	P	RIH directional BHA#8 on 5 7/8" DP f/ 226m to 1000m. ** Fill up string every 15 Std.

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Operations Summary

Start Time	End Time	Dur (hrs)	Phase	Code 1	Code 2	Time P-N-T	Comment
00:00	00:00		12 1/4"				Aug 22, 2014 00:00 - 01:30 Cont' RIH directional BHA#8 on 5 7/8" DP f/ 1000m to 1483m. Wash down to 1495m. 01:30 - 06:00 Fan bottom. Profile new bit and directional drilling 12 1/4" hole section in rotation mode f/ 1495m to 1533m w/ 700-880 gpm/ 900-1550 psi/ 5-30 kips/ 40-65 rpm/ 11-22 kft-lbs. >> Last sample @ 1530m: Anhy (80 %) + Dol (20 %)

Mud Information

Depth (mKB)	Type	Density (lb/gal)	Funnel Viscosity (s/qt)	PV Calc (cp)	YP Calc (lbf/100ft²)	Filtrate (mL/30min)
1,495.00	Polymer Mud	9.10	43	14.0	10.000	4.5
pH	**Pf (mL/mL)**	**Mf (mL/mL)**	**Pm (mL/mL)**	**Sand (%)**	**Low Gravity Solids (%)**	**High Gravity Solids (%)**
10.0	0.200	0.800	0.900			
Vis 3rpm	**Vis 6rpm**	**Percent Oil (%)**	**Percent Water (%)**	**Solids (%)**	**Chlorides (mg/L)**	**Filter Cake (/32")**
3	5	0.0	92.0	8.0	24,000.000	1
Gel 10 sec (lbf/100ft²)	**Gel 10 min (lbf/100ft²)**	**Gel 30 min (lbf/100ft²)**	**Oil Water Ratio**	**T (fl) (°C)**	**Active Mud Volume (bbl)**	**Reserve Mud Volume (bbl)**
4.000	6.000	9.000	0/100	55.0	1407.0	2085.0
Cum Mud Added (bbl)	**Mud Lost to Surface (bbl)**	**Mud Lost to Hole (bbl)**	**Avg Background Gas (%)**	**Avg Drill Gas (%)**	**K**	**n**
3044.0	23.0	0.0			8.037824	0.396

Marine Conditions

Swell Height (m)	Visibility (km)	Wind Direction (°)	Wind Speed (knots)	Barometric Pressure (mmHg)	Temperature - High (°C)
1.00	7.	245	12.0	996.0	35.0

Support Vessels

Vessel Name	Type	Arrival Date	Depart	Note
EP-HTN	Chopper	8/21/2014 11:26	8/21/2014 11:32	Pax in: 2 , Pax out: 3
Marina-1	Supply Boat	8/16/2014 15:15		C/Out Chopper duty. Stdby Anchorage for DD6 & Sahar 2,
Sea Command	Supply Boat	8/21/2014 16:45		Delivered cargo, B/L for Kish/Sharjah. Dep for Kish via Sahar-1. ETA Sahar-1 @ 0330 Hrs/22.08.2014.

Mud Additives

Description	Consumed	Cum Rec	On Loc
Caustic Soda 2014	0.0	244.0	224.0
Glydril	0.0	60.0	60.0
Duo-Tec	2.0	115.0	97.0
Pipe-Lax	0.0	18.0	18.0
KCL	0.0	37.0	37.0
Soda Ash 25 Kg	0.0	350.0	290.0
PTS (Temp. Stabilizer)	0.0	11.0	11.0
Sodium Bicarbonate	0.0	230.0	230.0
Starch	0.0	680.0	540.0

Bulk Amounts

Supply Item Description	Consumed	Received	Cum On Loc
Cement G	0.0	0.0	57.0
Diesel Fuel	8.0	0.0	139.0
Potable Water	20.0	17.0	113.0
Drill Water	28.0	0.0	313.0
Barite BB	0.0	0.0	0.0
Barite Bulk	0.0	0.0	249.0
Bentonite Bulk	0.0	0.0	8.0
Bentonite BB	0.0	0.0	0.0

Bit/BHA Information

BHA No.	Size (in)	Make	Model	Serial Number	Bit Type	IADC Codes	Nozzles (/32")
7	12 1/4	Varel	CH04JMRSV	255776	Insert	417	18/18/18/16
TFA (in²)	**Drilling Time (hrs)**	**Bit Revs**	**Depth In (mKB)**	**Depth Out (mKB)**	**Depth Drilled (m)**	**BHA ROP (m/hr)**	**IADC Bit Dull**
0.94	1.00	3,600.00	1,492.00	1,495.00	3.00	3.0	1-1-NO-A-E-I-NO-BHA
Flow Rate (min) (gpm)	**Flow Rate (max) (gpm)**	**SPP (min) (psi)**	**SPP (max) (psi)**	**WOB (min) (1000lbf)**	**WOB (max) (1000lbf)**	**RPM (min) (rpm)**	**RPM (max) (rpm)**
900	900	1,100.0	1,100.0	30	30	60	60
Circulating Time (hrs)	**Rotating Time (hrs)**	**% Rotating Time (%)**	**% Sliding Time (%)**	**PU HL (max) (1000lbf)**	**Rot HL (max) (1000lbf)**	**SO HL (min) (1000lbf)**	**String Wt (1000lbf)**
3.71	1.00	100.00	0.00				

Drilling Parameters

Depth Start (mKB)	Depth End (mKB)	Drilling Time (hrs)	Drilling Torque	Flow Rate (gpm)	RPM (rpm)	SPP (psi)	WOB (1000lbf)	Drill Str Wt (1000lbf)
1,492.00	1,495.00	1.00	18,000.0	900	60	1,100.0	30	

Drill String Components

Item Description	OD (in)	ID (in)	Len (m)	Cum Len (m)
15 jts 5 7/8" HWDP	5 7/8	4.250	140.00	172.25
X/O	8	2.810	0.79	32.25
3 jts 8 1/4" DC	8 1/4	2.810	28.01	31.46
12 1/8" Stabilizer (COR-3057-AD)	12 1/8	2.810	2.18	3.45
Bit sub with float valve inside	9 1/2		0.95	1.27
12 1/4" Insert Bit-RR	12 1/4		0.32	0.32

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Bit/BHA Information

BHA No.	Size (in)	Make	Model	Serial Number	Bit Type	IADC Codes	Nozzles (/32")
8	12 1/4	BEST	M1385	228378	PDC	M432	12/12/12/12/13/13/13/13
TFA (in²)	**Drilling Time (hrs)**	**Bit Revs**	**Depth In (mKB)**	**Depth Out (mKB)**	**Depth Drilled (m)**	**BHA ROP (m/hr)**	**IADC Bit Dull**
0.95	0.00						-------
Flow Rate (min) (gpm)	**Flow Rate (max) (gpm)**	**SPP (min) (psi)**	**SPP (max) (psi)**	**WOB (min) (1000lbf)**	**WOB (max) (1000lbf)**	**RPM (min) (rpm)**	**RPM (max) (rpm)**
Circulating Time (hrs)	**Rotating Time (hrs)**	**% Rotating Time (%)**	**% Sliding Time (%)**	**PU HL (max) (1000lbf)**	**Rot HL (max) (1000lbf)**	**SO HL (min) (1000lbf)**	**String Wt (1000lbf)**
0.15							

Drilling Parameters

Depth Start (mKB)	Depth End (mKB)	Drilling Time (hrs)	Drilling Torque	Flow Rate (gpm)	RPM (rpm)	SPP (psi)	WOB (1000lbf)	Drill Str Wt (1000lbf)

Drill String Components

Item Description	OD (in)	ID (in)	Len (m)	Cum Len (m)
3 jts 5 7/8" HWDP	5 7/8	4.250	27.83	226.57
7" Drilling Jar (COR-3019-AD)	7	3.000	5.85	198.74
16 jts 5 7/8" HWDP	5 7/8	4.250	149.23	192.89
X/O	7	2.810	0.79	43.66
1 jt 8 1/4" DC	8 1/4	2.810	9.46	42.87
8 1/2" Float Sub	8 1/2		0.79	33.41
8" NMDC	8	3.500	9.44	32.62
8" TM - Sub	8		2.03	23.18
8" NMDC	8	3.500	9.46	21.15
12" S. Stabilizer (COR3059AD)	11 7/8	3.125	2.19	11.69
9 5/8" Motor (AKO: 1.00° , Sleeve: 12 1/8")	9 5/8		9.14	9.50
12 1/4" PDC bit	12 1/4		0.36	0.36

Well Control

No.	Depth (mKB)	Stroke (in)	Strokes (spm)	Pres (psi)	Eff (%)	Q (gpm)

Test Date	Last Casing String Run	Test Type	Depth (mKB)	TVD (mKB)	Applied Surface Pressure (psi)
4/8/2014	18 5/8" Shoe, 870.00mKB	Formation Integrity	870.00	843.25	400.0
Applied Surface Pressure (psi)	**Volume Pumped (bbl)**	**Current MAASP**	**Kick Tolerance**	**Dens Fluid (lb/gal)**	**Leak Off Pressure (psi)**
400.0	8.0	400.00		11.48	1,650.4
Test Date	**Last Casing String Run**	**Test Type**	**Depth (mKB)**	**TVD (mKB)**	**Applied Surface Pressure (psi)**
8/21/2014	13 3/8" Shoe, 1,487.00mKB	Formation Integrity	1,495.00	1,427.34	1,200.0
Applied Surface Pressure (psi)	**Volume Pumped (bbl)**	**Current MAASP**	**Kick Tolerance**	**Dens Fluid (lb/gal)**	**Leak Off Pressure (psi)**
1,200.0	7.7	1,200.00		14.03	3,413.7

Geology Information

Formation Name	Prog Top MD (mKB)	Final Top MD (mKB)	Final Top TVD (mKB)
Upper Gadvan		1,209.00	1,156.52
Khalij		1,231.00	1,176.98
Lower Gadvan		1,247.00	1,191.88
Fahliyan		1,286.50	1,228.87
Hith		1,469.30	1,402.67

General Remarks

Comment

>>> Requirements: (Outstanding of MI MR#12: 2 Pallets of Pac-LV) + (150 MT Cement) + (7" Jar) + (100 MT FO).

>>> Sent:
** Sea Command to Kish: (Mehran Cementing:Basket contains 2 x Swage + X/O and Meharan Equipment + 2 x Skid of 10 3/4" Double Plug CMT Head w/ Related X/O + Recirculating Pump) + (Aban Equipment) + (HPOGC: 9 5/8" Motor) + (Garbage).

>>> Received:
** Sea Command f/ Kish: (MI Chemicals: 20 BB KCl + 96 Can Caustic Soda + 8 Drum Detergent + 40 Drum Gly Drill + 12 Drum Pipe Lax + 8 Drum PTS200 + 200 Sxs Soda Ash + 80 Sxs Sodium Bicarbonate + 400 Sxs Starch) + (HPOGC: 9 5/8" Motor) + (Al-Masood H2S Equipment) + (COR: 12" String Stab).